

Supplementary Materials for

Persistence of mature dendritic cells, T_H2A, and Tc2 cells characterize clinically resolved atopic dermatitis under IL-4R α blockade

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Published 22 January 2021, *Sci. Immunol.* **6**, eabe2749 (2021)
DOI: 10.1126/sciimmunol.abe2749

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Other Supplementary Material for this manuscript includes the following:

(available at immunology.sciencemag.org/cgi/content/full/6/55/eabe2749/DC1)

Table S5 (Microsoft Excel format). Differential gene expression analysis within cell types of scRNA-seq analyses; comparison of healthy controls versus AD untreated.

Table S6 (Microsoft Excel format). Differential gene expression analysis within cell types of scRNA-seq analyses; comparison of 16 weeks dupilumab-treated AD versus AD untreated.

Table S7 (Microsoft Excel format). Differential gene expression analysis within cell types of scRNA-seq analyses; comparison of 1 year dupilumab-treated AD versus AD untreated.

Table S8 (Microsoft Excel format). Differential gene expression analysis within cell types of scRNA-seq analyses; comparison of 16 weeks of dupilumab-treated AD versus healthy controls.

Table S9 (Microsoft Excel format). Differential gene expression analysis within cell types of scRNA-seq analyses; comparison of 1 year of dupilumab-treated AD versus healthy controls.

Table S10 (Microsoft Excel format). Differential gene expression analysis within cell types of scRNA-seq analyses; comparison of 1 year of dupilumab-treated AD versus 16 weeks of dupilumab-treated AD.

Table S12 (Microsoft Excel format). Differential protein expression of proteomic multiplex assays from suction blister fluid samples.

Table S14 (Microsoft Excel format). Differential protein expression of proteomic multiplex assays from serum samples.

Table S15. Raw data file (Excel spreadsheet).

SUPPLEMENTARY FIGURES

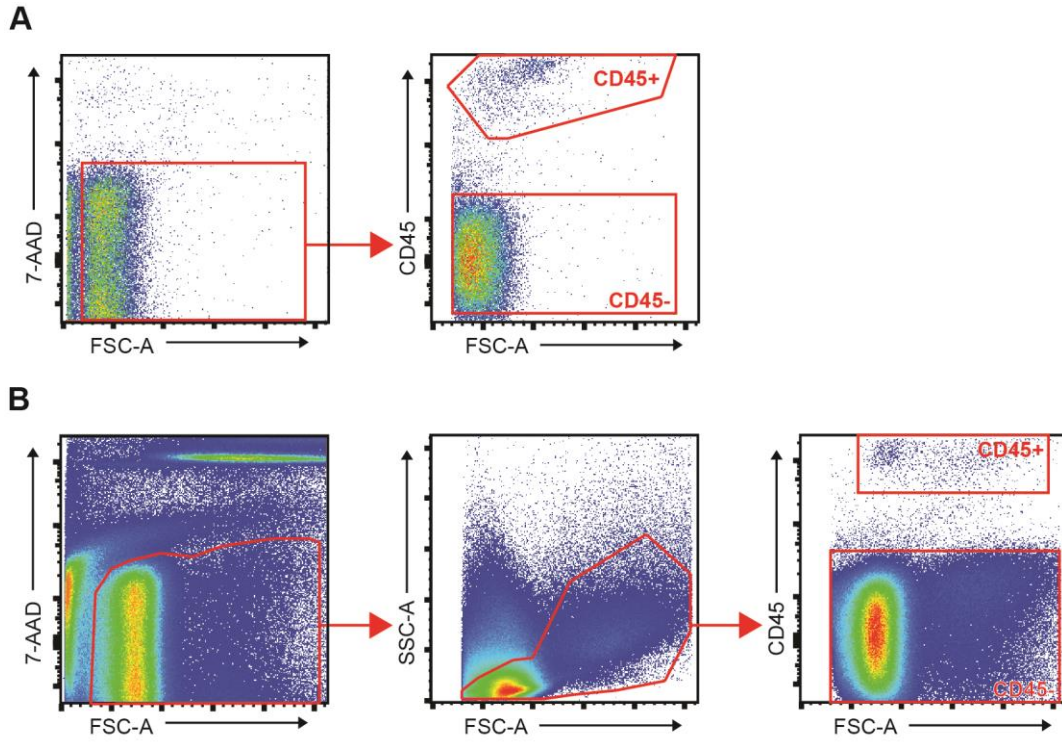


Fig. S1. Representative plots from FACS. (A) blister fluid cells and **(B)** digested epidermal sheets (i.e. the blister roof) from suction blister samples. CD45+ and CD45- cells were sorted and subsequently pooled at equal amounts for scRNA-seq analyses.

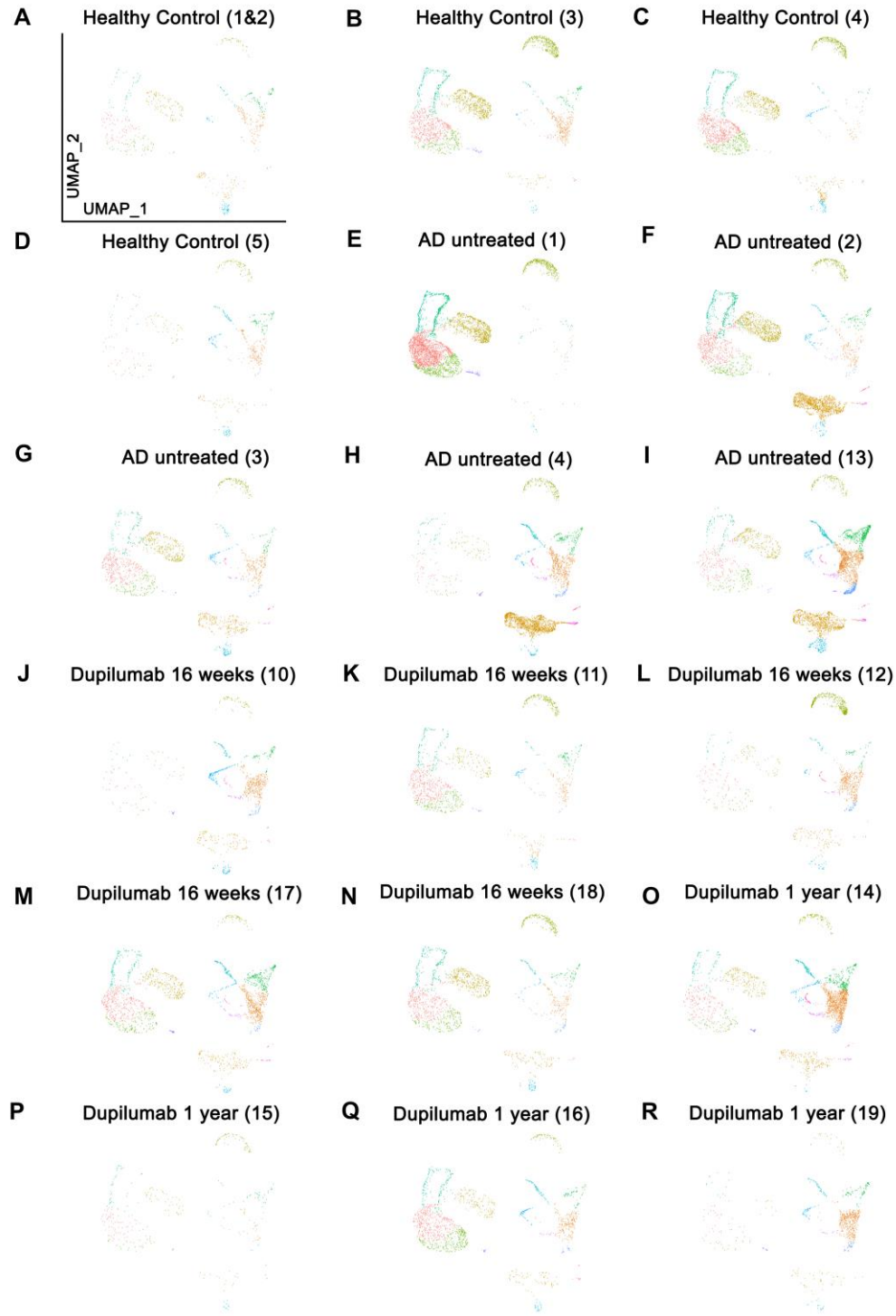


Fig. S2. scRNA-seq map of individual suction blister donors. (A-R) UMAP plots of samples as used for all single-cell analyses. Numbers in brackets correspond to sample ID of atopic dermatitis (AD) and healthy control (HC) individuals listed in Table S1.

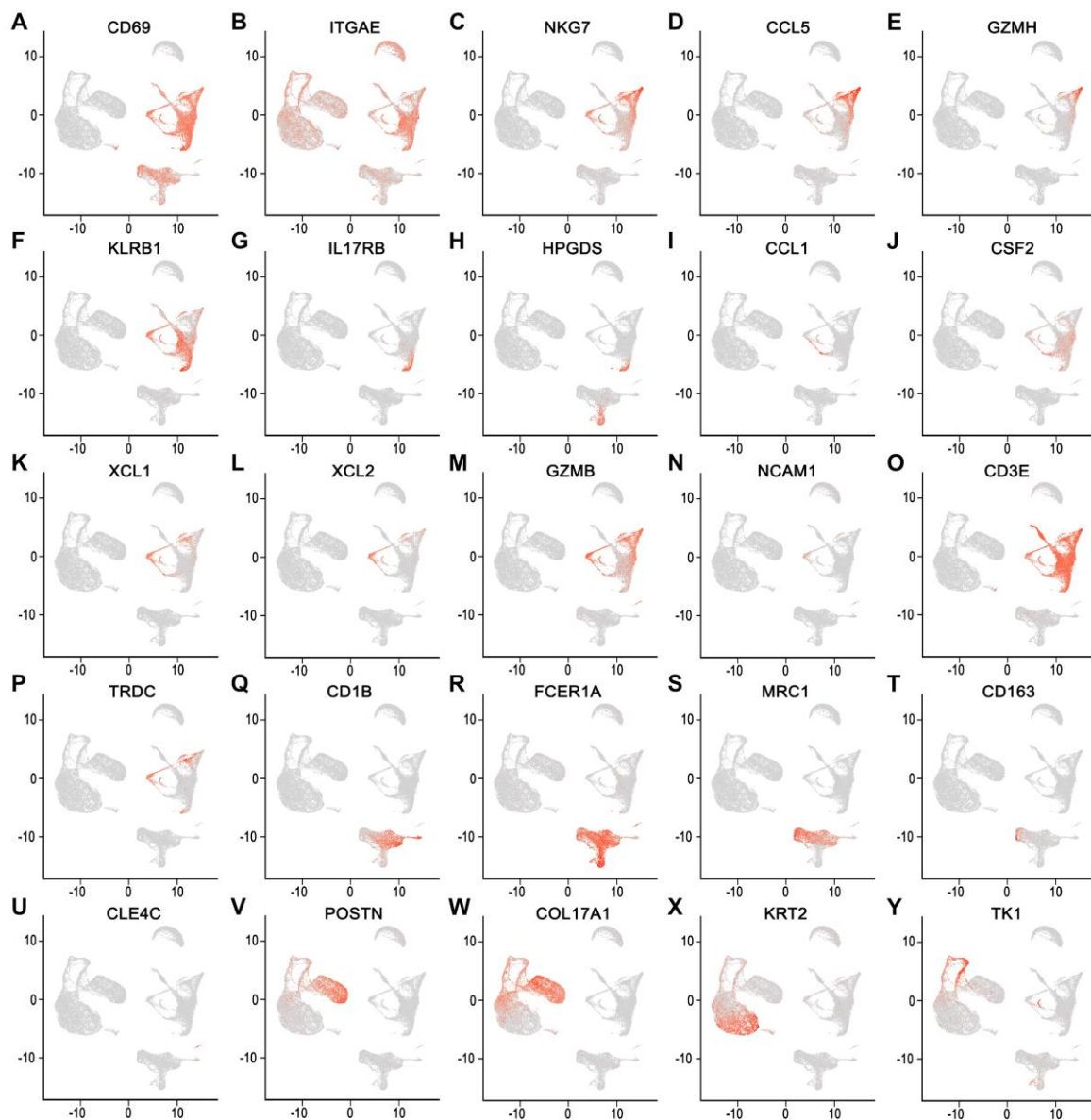


Fig. S3. Feature plots of selected marker genes for the entire scRNA-seq dataset. (A-Y) Expression levels for each cell are color-coded (red) and overlaid onto UMAP plots. Intensity of color reflects respective level of expression.

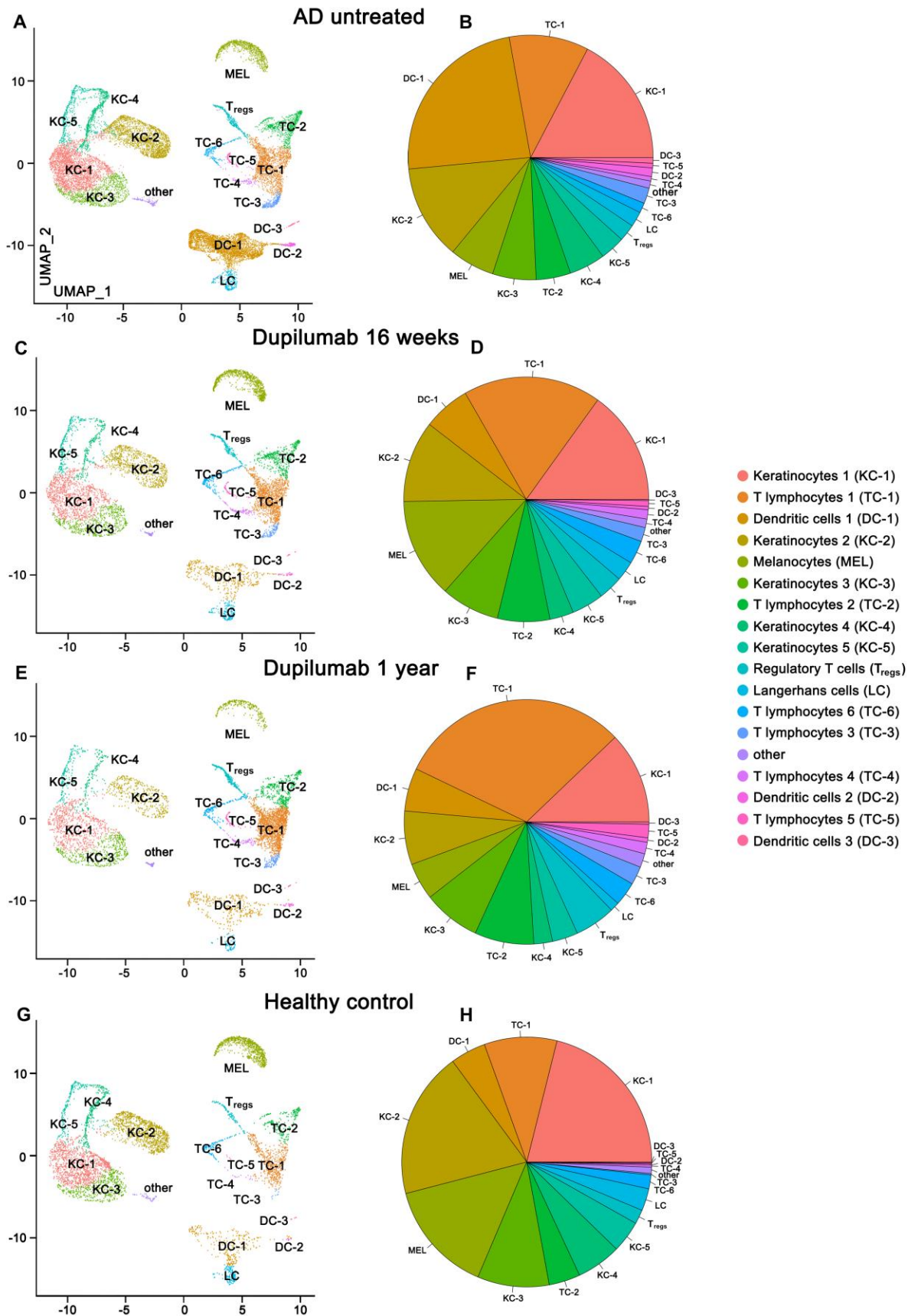


Fig. S4. scRNA-seq map of cell populations in untreated AD, dupilumab-treated AD, and healthy control suction blisters. (A-B) untreated atopic dermatitis (n=16,974 cells), **(C-D)** 16 weeks of dupilumab treatment (n=7,575), **(E-F)** one year of dupilumab treatment (n=5,565), and **(G-H)** healthy control blister samples (n=5,554). Color-coded frequencies of cell clusters for depicted groups are displayed in pie graphs (B, D, F, H). AD: atopic dermatitis.

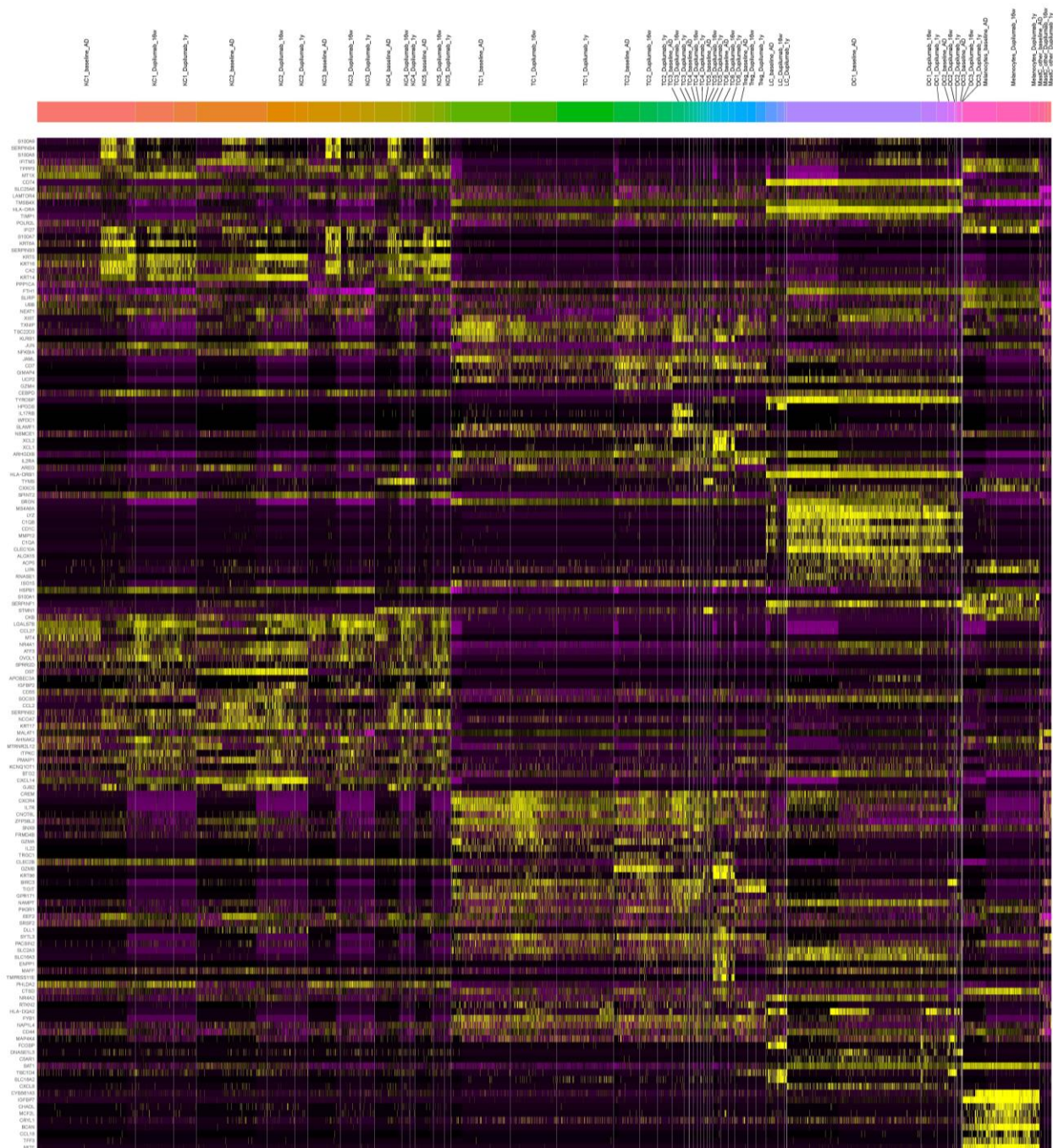


Fig. S5. Expression of down-regulated genes in dupilumab-treated AD skin. Expression heatmap showing the top 10 downregulated differentially expressed genes per cell cluster, sorted according to highest log fold change ordered by smallest adjusted p-value using logistic regression with Bonferroni correction. Comparisons of 16 weeks dupilumab vs baseline

(untreated AD), and 1 year vs 16 weeks of dupilumab treatment. Yellow color denotes high, and purple color denotes low expression. AD: atopic dermatitis; 16w: 16 weeks of dupilumab; 1y: 1 year of dupilumab.

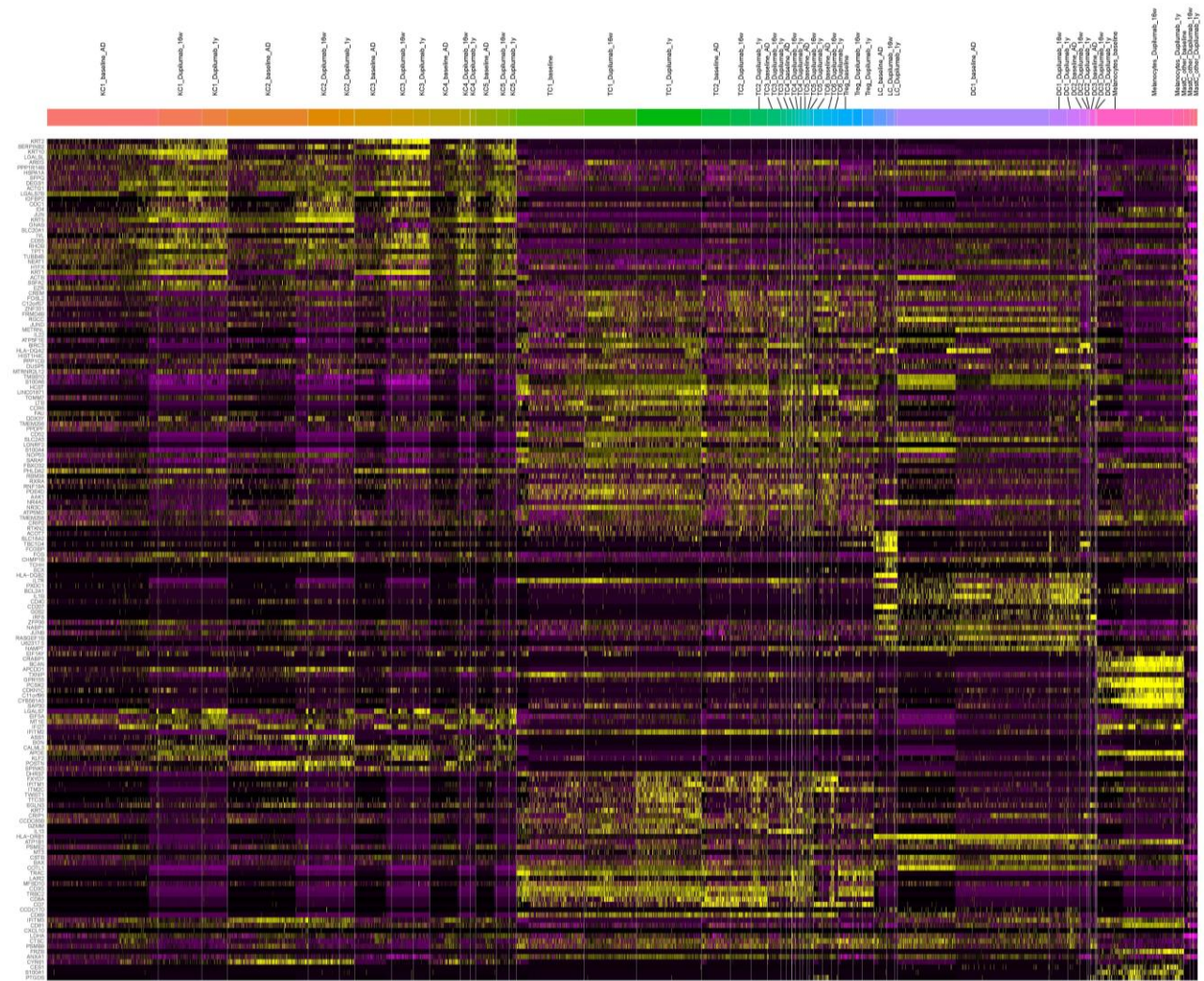


Fig. S6. Expression of up-regulated genes in dupilumab-treated AD skin. Expression heatmap showing the top 10 downregulated differentially expressed genes per cell cluster, sorted according to highest log fold change ordered by smallest adjusted p-value using logistic regression with Bonferroni correction. Comparisons of 16 weeks dupilumab vs baseline (untreated AD), and 1 year vs 16 weeks of dupilumab treatment. Yellow color denotes high, and purple color denotes low expression. AD: atopic dermatitis; 16w: 16 weeks of dupilumab; 1y: 1 year of dupilumab.

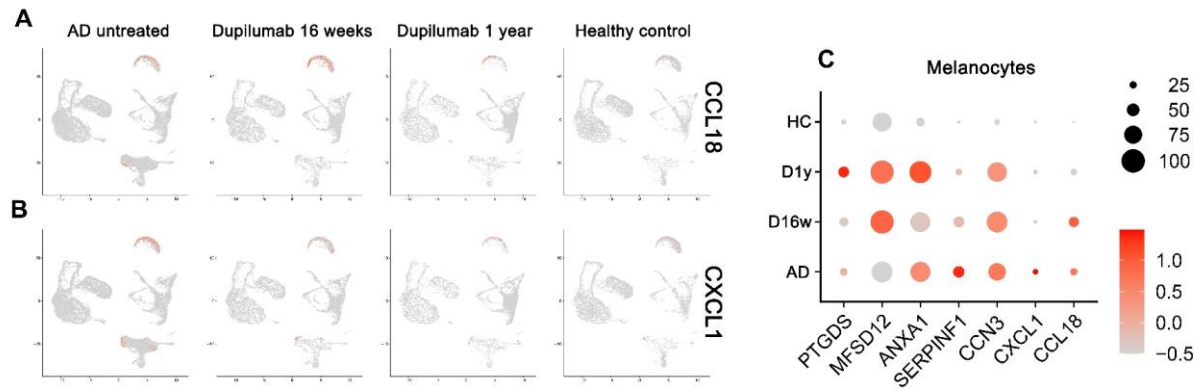


Fig. S7. Gene expression within the melanocyte cluster. Feature plots of **(A)** CCL18 and **(B)** CXCL1 expression distribution in each sample group. **(C)** Gene expression dot plots of melanocytes displaying average (red color) and frequency of expression (circle size) of selected melanocyte marker genes in each sample group. AD: Atopic dermatitis; D16w: 16 weeks of dupilumab-treated AD; D1y: 1 year of dupilumab-treated AD; HC: healthy control.

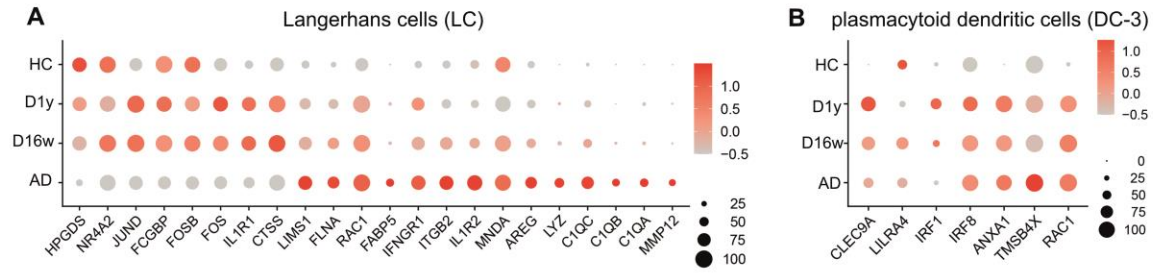


Fig. S8. Gene expression within LC and plasmacytoid DC clusters. (A) LC and **(B)** plasmacytoid DC (pDC, DC-3) clusters delineating maturation, pro- and anti-inflammatory markers of interest by gene expression dot plots. Average expression is reflected by intensity of red color, cell frequency by circle size. AD: atopic dermatitis; D16w: 16 weeks of dupilumab-treated AD; D1y: 1 year of dupilumab-treated AD; HC: healthy control

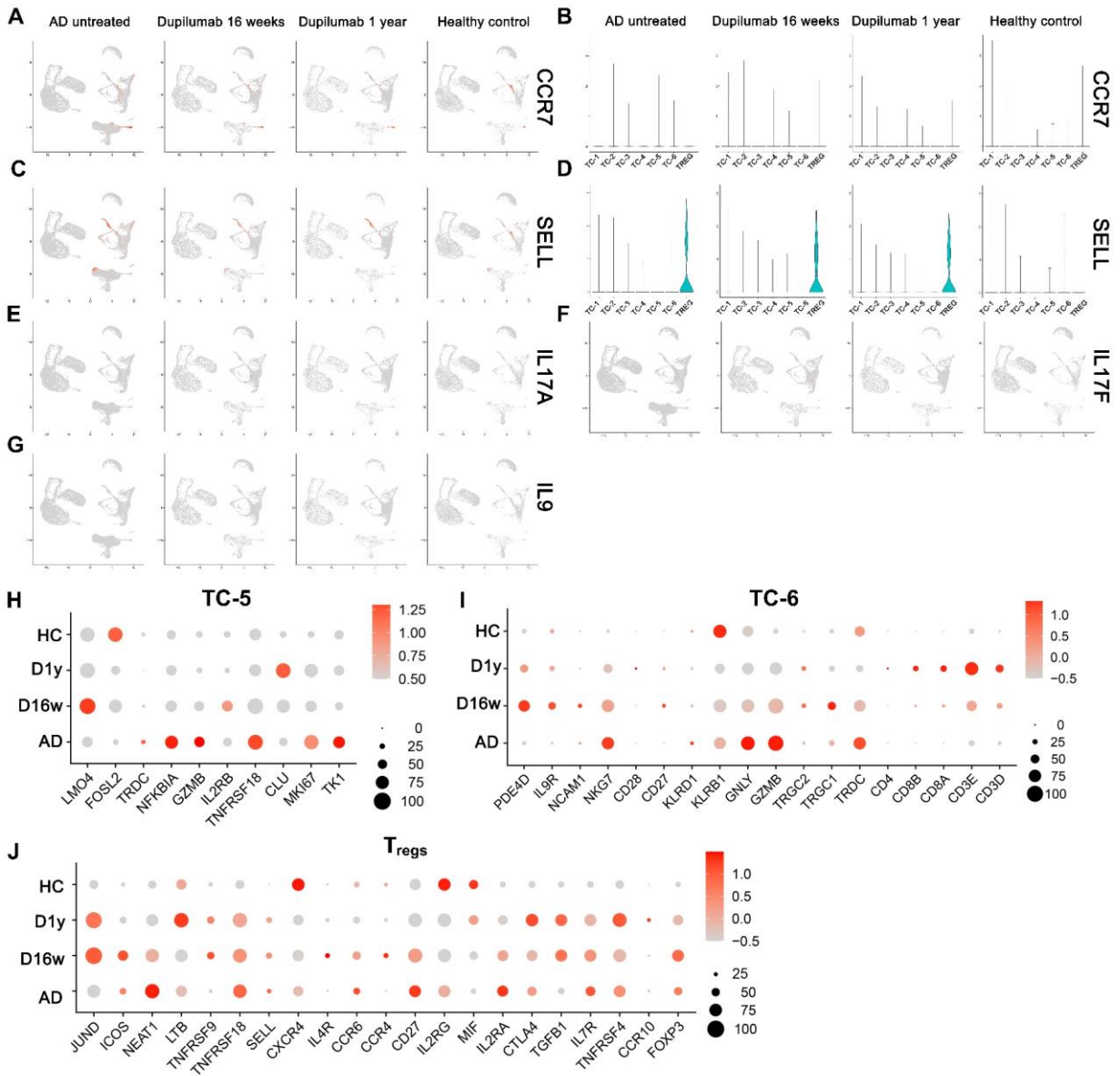


Fig. S9. Selected T cell marker gene expression. scRNA-seq as depicted by feature plots (A,C,E-G), violin plots (B,D) and dot plots (H-J). Dots in feature plots represent individual cells, and red denotes respective gene expression overlaid onto UMAP plots. Violin plots show the distribution of normalized expression levels of selected genes. Gene expression dot plots of selected markers of denoted T-cell clusters. Average expression is reflected by intensity of red color, frequency by circle size. AD: atopic dermatitis; D16w: 16 weeks of dupilumab-treated AD; D1y: 1 year of dupilumab-treated AD; HC: healthy control

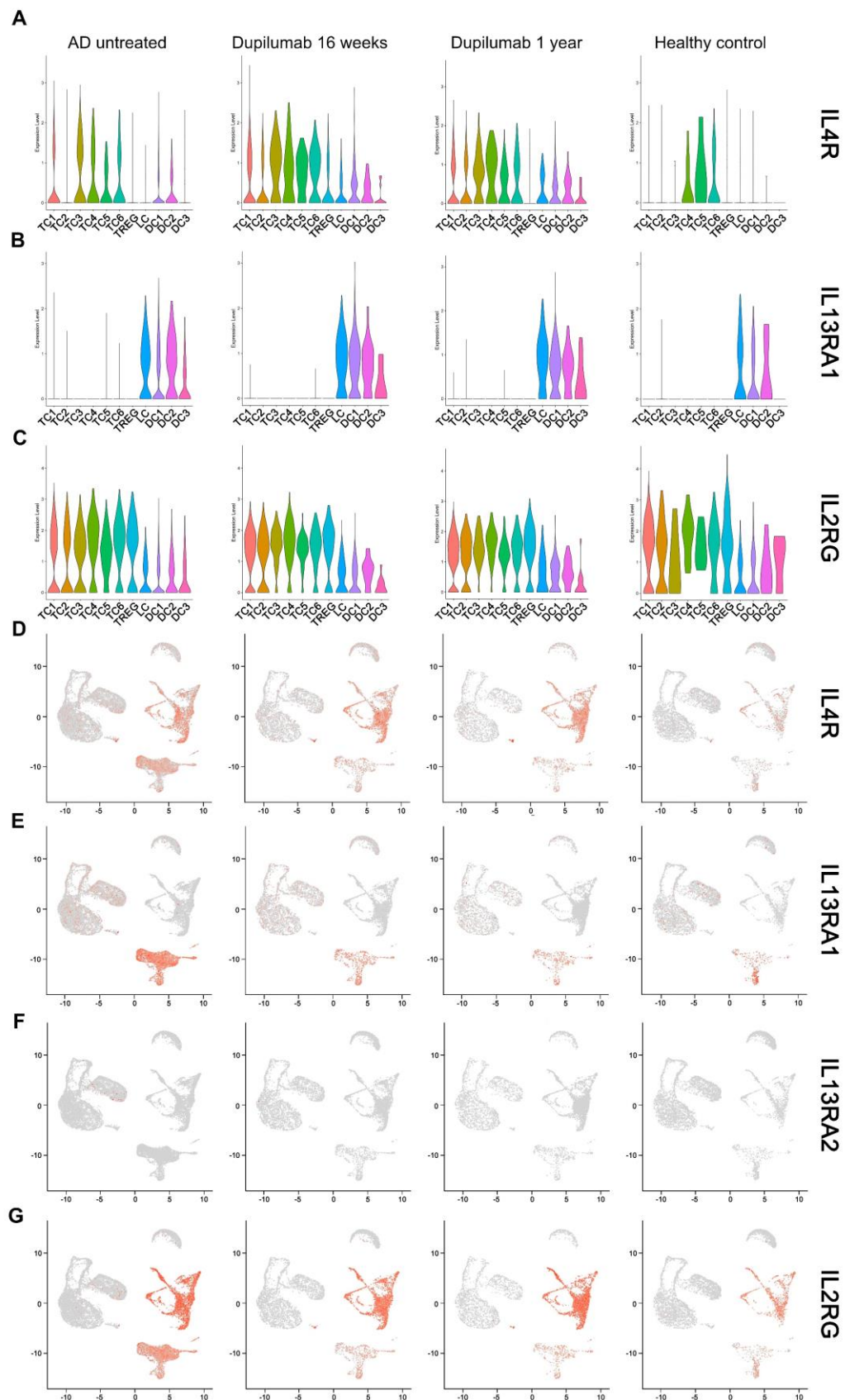


Fig. S10. Expression of receptor components for IL-4 and IL-13 in T cell and DC clusters.

Violin plots and feature plots of **(A, D)** IL4R **(B, E)** IL13RA1 **(F)** IL13RA2 and **(C, G)** IL2RG sorted per sample group showing a distinct receptor expression profile in T-cell and DC clusters. Violin plots show the distribution of normalized expression levels of selected genes; dots represent individual cells, and red denotes respective gene expression overlaid onto UMAP plots.

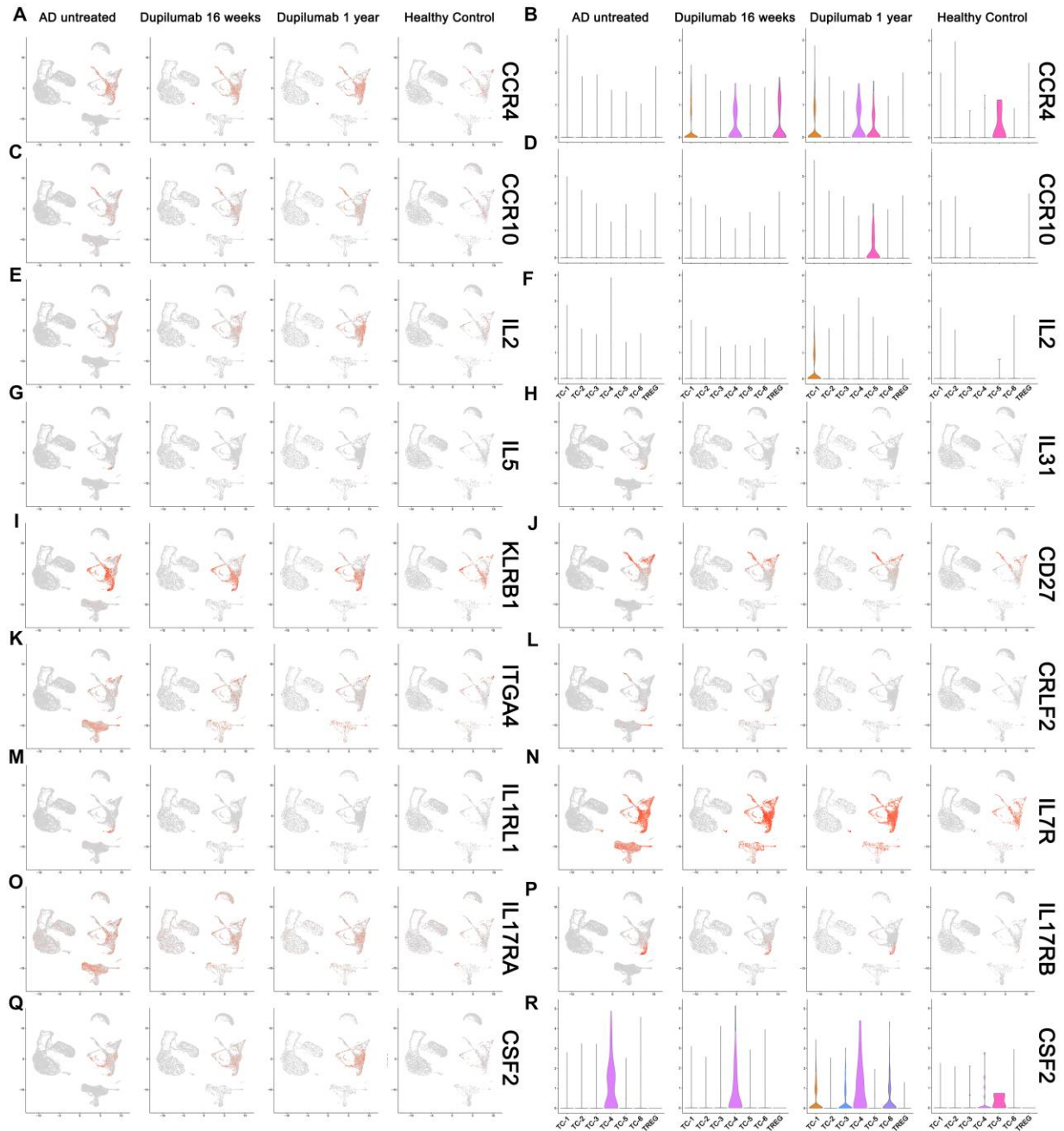


Fig. S11. Selected immune cell markers. scRNA-seq feature plots (red denotes respective gene expression) and violin plots showing expression levels (y axis) in the respective cluster (**A-R**). Violin plots show the distribution of normalized expression levels of selected genes; dots represent individual cells, and red denotes respective gene expression overlaid onto UMAP plots.

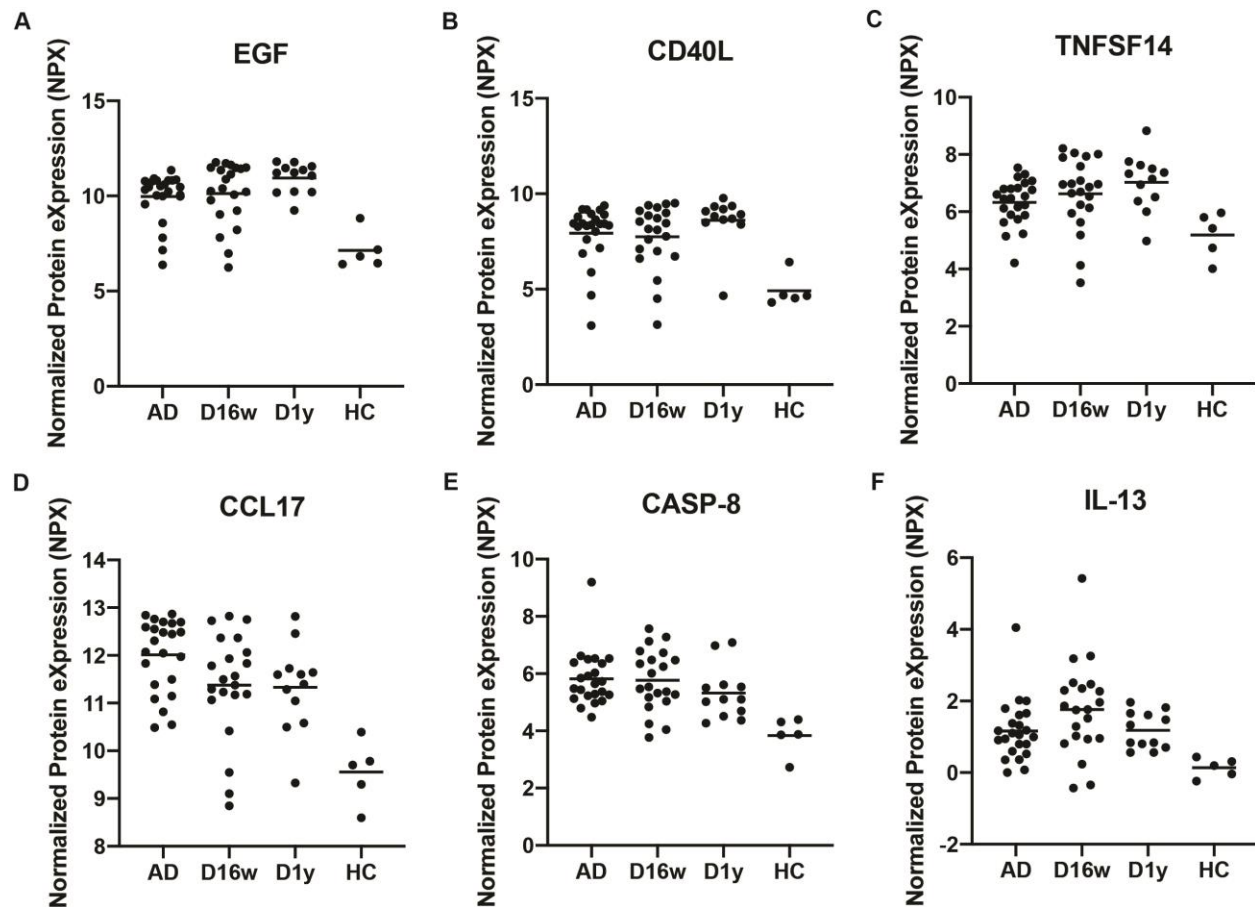


Fig. S12. Log2 NPX of individual serum samples for markers significantly increased at 1 year of dupilumab treatment compared with healthy control individuals for all time points. The markers depicted are those that were significantly increased at the 1 year dupilumab time point compared to healthy controls (at a $\log_2 \text{FCH} > |1.0|$ and adjusted $p\text{-value} < 0.05$), as depicted in Figure 7C. Horizontal bars indicate means. AD: Atopic dermatitis; D16w: 16 weeks of dupilumab-treated AD; D1y: one year of dupilumab-treated AD; HC: Healthy controls.

SUPPLEMENTARY TABLES

Table S1. Baseline characteristics of patients and healthy controls included for skin suction blistering and skin biopsy

sampling. FU: follow-up; AD: Atopic dermatitis; EASI: Eczema Area and Severity Index; HC: Healthy control; n.a. not available.

scRNAseq samples can be accessed via GEO identifier GSE158432 or GSE153760. scRNAseq and proteomic samples listed in

GSE153760 (AD1, AD2, AD3, AD4, and HC samples) have previously been published in Rojahn et al (PMID 32344053) and were

re-analyzed for this publication.

ID	Age (y)	Sex	Race	EASI (% reduction from baseline)	Diagnosis	Group	Total serum IgE (kU/L)	Blood Eosinophils (%)	Sample type	Multiplex proteomics	scRNAseq (GEO Identifier)	IHC	qPCR
AD1	18	male	Caucasian	34.2 (0)	Atopic dermatitis	untreated	>5000	14	Suction blister	x	GSM4653855		
AD2	19	female	Caucasian	44.6 (0)	Atopic dermatitis	untreated	8.7	5.8	Suction blister	x	GSM4653856		
AD3	34	male	Caucasian	44.7 (0)	Atopic dermatitis	untreated	1138	9.6	Suction blister	x	GSM4653857		
AD4	33	female	Caucasian	5.5 (0)	Atopic dermatitis	untreated	180	4.2	Suction blister	x	GSM4653858		
AD13	45	female	Caucasian	27.4 (0)	Atopic dermatitis	untreated	n.a.	9	Suction blister	x	GSM4800173		
AD31	30	female	Caucasian	24.1 (0)	Atopic dermatitis	untreated	n.a.	n.a.	Suction blister	x			
AD10 (FU of AD3)	35	male	Caucasian	15.9 (64.4)	Atopic dermatitis	Dupilumab 16 weeks	169	5.7	Suction blister	x	GSM4800170		
AD11 (FU of AD1)	19	male	Caucasian	10.1 (70.5)	Atopic dermatitis	Dupilumab 16 weeks	1702	13	Suction blister	x	GSM4800171		
AD12 (FU of AD2)	19	female	Caucasian	2.4 (94.6)	Atopic dermatitis	Dupilumab 16 weeks	n.a.	n.a.	Suction blister	x	GSM4800172		
AD17	41	male	Caucasian	9 (50.0)	Atopic dermatitis	Dupilumab 16 weeks	579	3.5	Suction blister	x	GSM4800177		
AD18 (FU of AD13)	45	female	Caucasian	5.4 (80.3)	Atopic dermatitis	Dupilumab 16 weeks	2.82	2.6	Suction blister	x	GSM4800178		
AD14	60	female	Caucasian	3.2 (92.9)	Atopic dermatitis	Dupilumab 1 year	3729	21	Suction blister	x	GSM4800174		
AD15	28	female	Caucasian	2.6 (90.0)	Atopic dermatitis	Dupilumab 1 year	118	7	Suction blister	x	GSM4800175		

AD16	41	male	Caucasian	1.1 (97.4)	Atopic dermatitis	Dupilumab 1 year	1097	6.4	Suction blister	x	GSM4800176		
AD19	46	male	Caucasian	11.2 (74.8)	Atopic dermatitis	Dupilumab 1 year	1041	8	Suction blister	x	GSM4800179		
HC1&2	42	female	Caucasian	n.a.	Healthy control	n.a.	n.a.	n.a.	Suction blister	x	GSM4653863 GSM4653864		
HC3	47	female	Caucasian	n.a.	Healthy control	n.a.	n.a.	n.a.	Suction blister	x	GSM4653865		
HC4	49	female	Caucasian	n.a.	Healthy control	n.a.	n.a.	n.a.	Suction blister	x	GSM4653866		
HC5	39	male	Caucasian	n.a.	Healthy control	n.a.	n.a.	n.a.	Suction blister	x	GSM4653867		
HC6	40	male	Caucasian	n.a.	Healthy control	n.a.	n.a.	n.a.	Suction blister	x			
HC7	40	male	Caucasian	n.a.	Healthy control	n.a.	n.a.	n.a.	Suction blister	x			
HC8	28	female	Caucasian	n.a.	Healthy control	n.a.	n.a.	n.a.	Suction blister	x			
HC9	44	female	Caucasian	n.a.	Healthy control	n.a.	n.a.	n.a.	Skin biopsy			x	x
HC10	52	female	Caucasian	n.a.	Healthy control	n.a.	n.a.	n.a.	Skin biopsy			x	x
HC11	38	female	Caucasian	n.a.	Healthy control	n.a.	n.a.	n.a.	Skin biopsy			x	x
HC12	52	male	Caucasian	n.a.	Healthy control	n.a.	n.a.	n.a.	Skin biopsy			x	x
HC13	34	female	Caucasian	n.a.	Healthy control	n.a.	n.a.	n.a.	Skin biopsy			x	x
AD21	52	female	Caucasian	27.4 (0)	Atopic dermatitis	untreated	160	8.9	Skin biopsy			x	x
AD22	19	female	Caucasian	6.8 (0)	Atopic dermatitis	untreated	65.7	5	Skin biopsy			x	x
AD23	28	male	Caucasian	n.a.	Atopic dermatitis	untreated	4452	n.a.	Skin biopsy			x	x
AD24	50	male	Caucasian	n.a.	Atopic dermatitis	untreated	>5000	8.2	Skin biopsy			x	x
AD25	33	male	Caucasian	10.7 (68.2)	Atopic dermatitis	Dupilumab 1 year	n.a.	n.a.	Skin biopsy			x	x
AD26	55	female	Caucasian	15.1 (71.0)	Atopic dermatitis	Dupilumab 1 year	>5000	12	Skin biopsy			x	x
AD27	30	male	Caucasian	2.1 (95.8)	Atopic dermatitis	Dupilumab 1 year	3862	n.a.	Skin biopsy			x	x
AD28	40	male	Caucasian	15.7 (69.5)	Atopic dermatitis	Dupilumab 1 year	1716	3.4	Skin biopsy			x	x
AD29	55	female	Caucasian	0 (100)	Atopic dermatitis	Dupilumab 1 year	1204	1.5	Skin biopsy			x	x

Table S2. Cell numbers, read numbers, and gene numbers. For each suction blister sample used for single cell RNA sequencing (unfiltered counts before quality control). AD: Atopic dermatitis; HC: Healthy control.

Sample ID	Cell number	Read number	Gene number
AD1	4864	17160	1158
AD2	4104	16031	1784
AD3	2084	36272	1994
AD4	3484	22785	2179
AD10	1217	67808	2746
AD11	2037	44324	2515
AD12	1814	46110	3000
AD13	5327	14870	1464
AD14	3701	21710	1803
AD15	756	96559	1604
AD16	3673	22498	241
AD17	3984	20019	910
AD18	3066	25507	1261
AD19	1060	77992	2331
HC1	171	509171	273
HC2	751	125639	1024
HC3	2709	31222	1206
HC4	2277	35237	1520
HC5	915	106691	1936

Table S3. Cell numbers per patient and cell cluster. Absolute cell numbers per patient and cell clusters after quality control filtering, as used for figures and all calculations based on scRNA-seq.

	KC-1	KC-2	KC-3	KC-4	KC-5	TC-1	TC-2	TC-3	TC-4	TC-5	TC-6	Treg	LC	DC-1	DC-2	DC-3	Melanocytes	MastC_other
AD1	1903	781	525	388	257	16	10	1	0	0	9	2	9	11	0	0	376	85
AD2	474	710	187	268	179	128	82	13	11	8	55	37	46	1286	28	26	196	6
AD3	316	308	127	76	57	181	46	15	19	20	62	24	71	243	8	30	119	9
AD4	50	58	26	22	27	473	166	53	27	40	47	121	50	1560	98	48	181	25
AD10	24	21	18	4	10	340	75	33	25	16	131	52	55	106	5	3	55	16
AD11	353	123	179	51	84	86	73	6	10	1	31	14	42	50	5	0	196	13
AD12	66	56	42	11	13	371	52	52	18	27	13	29	27	91	4	1	479	2
AD13	196	268	110	56	35	984	471	267	43	35	54	197	163	923	60	24	136	36
AD14	170	201	88	36	64	1044	258	69	52	92	68	167	11	194	18	12	115	55
AD15	70	44	29	8	35	25	28	4	5	1	6	4	11	15	2	2	66	4
AD16	410	139	280	89	81	198	61	20	14	3	56	75	29	66	11	3	70	21
AD17	345	243	166	67	104	468	258	24	30	13	43	106	28	130	16	8	56	47
AD18	354	373	180	106	111	115	65	27	10	1	16	42	45	94	3	0	214	7
AD19	26	10	25	5	9	443	88	42	14	3	45	53	17	42	2	0	21	21
HC1&2	81	103	56	28	15	113	72	3	0	0	12	19	50	67	0	4	34	5
HC3	488	536	190	150	104	244	65	1	7	2	13	25	14	37	3	0	403	28
HC4	578	375	248	134	98	27	18	0	4	0	38	13	59	108	3	0	294	7
HC5	25	40	21	10	11	138	70	6	5	4	37	45	36	43	3	2	77	5

Table S4. Top 10 differentially expressed genes (Clustermarker). According to highest log fold change ordered by smallest adjusted p-value using logistic regression with Bonferroni correction for each cluster, as compared to the rest of the dataset in all sequenced skin cells.

p_val	avg_logFC	pct.1	pct.2	p_val_adj	Cluster #	Gene	Cluster Name
0	1.562441	0.994	0.344	0	0	LY6D	KC-1
0	1.559729	0.994	0.364	0	0	DMKN	
0	1.554234	0.976	0.515	0	0	MT1X	
0	1.550577	0.988	0.353	0	0	KRT1	
0	1.462539	1	0.423	0	0	SFN	
0	1.360942	0.994	0.734	0	0	KRT10	
0	1.338991	0.984	0.334	0	0	DSP	
0	1.303997	0.979	0.326	0	0	S100A14	
0	1.283674	0.915	0.261	0	0	LYPD3	
0	1.273764	0.962	0.309	0	0	LGALS7B	
0	1.781945	0.991	0.195	0	1	IL32	TC-1
0	1.527208	0.94	0.181	0	1	TRAC	
0	1.491727	0.897	0.212	0	1	IL7R	
0	1.459129	0.261	0.017	0	1	IL22	
0	1.436706	0.921	0.269	0	1	CXCR4	
0	1.43364	0.786	0.174	0	1	NBAS	
0	1.413875	0.943	0.14	0	1	CD3D	
0	1.390928	0.838	0.136	0	1	TRBC2	
0	1.382279	0.966	0.317	0	1	CD52	
0	1.35879	0.935	0.259	0	1	ALOX5AP	
0	3.411777	0.949	0.04	0	2	LYZ	DC-1
0	3.264058	0.995	0.246	0	2	HLA-DRA	
0	2.957326	0.992	0.356	0	2	HLA-DRB1	

0	2.910716	0.472	0.02	0	2	MMP12	
0	2.900969	0.99	0.312	0	2	HLA-DPA1	
0	2.849953	0.991	0.279	0	2	HLA-DPB1	
0	2.770454	0.958	0.178	0	2	HLA-DQA1	
0	2.742218	0.993	0.568	0	2	CST3	
0	2.729523	0.618	0.017	0	2	IL1B	
0	2.728513	0.996	0.376	0	2	CD74	
0	2.84626	0.841	0.049	0	3	KRT15	KC-2
0	2.311265	0.768	0.105	0	3	CYR61	
0	2.270851	0.995	0.358	0	3	CXCL14	
0	2.051747	0.94	0.273	0	3	DST	
0	1.993852	1	0.544	0	3	KRT14	
0	1.798266	0.882	0.131	0	3	COL17A1	
0	1.638649	0.989	0.323	0	3	S100A2	
0	1.605884	0.694	0.046	0	3	POSTN	
0	1.605067	0.379	0.025	0	3	CCL2	
0	1.549659	1	0.398	0	3	KRT5	
0	4.792368	0.995	0.026	0	4	DCT	Melanocytes
0	4.595875	0.986	0.039	0	4	TYRP1	
0	4.561629	0.989	0.047	0	4	PMEL	
0	4.186377	0.994	0.029	0	4	MLANA	
0	3.890764	0.991	0.088	0	4	IGFBP7	
0	2.909147	0.962	0.007	0	4	APOD	
0	2.89395	0.959	0.016	0	4	MITF	
0	2.874788	0.987	0.15	0	4	QPCT	
0	2.658301	0.98	0.03	0	4	GPM6B	
0	2.634864	0.993	0.413	0	4	CD59	
0	3.326048	0.803	0.097	0	5	KRT2	KC-3
0	2.763954	0.995	0.263	0	5	KRTDAP	
0	2.236822	1	0.76	0	5	KRT10	

0	2.201317	0.965	0.184	0	5	SBSN	
0	2.084868	0.999	0.418	0	5	KRT1	
0	1.997979	0.82	0.142	0	5	CALML5	
0	1.856314	0.971	0.324	0	5	LYPD3	
0	1.832921	1	0.429	0	5	DMKN	
0	1.724433	0.543	0.112	0	5	SPRR1B	
0	1.659911	0.224	0.044	0	5	S100A7	
0	3.040767	0.824	0.03	0	6	CCL5	TC-2
0	2.447717	0.818	0.065	0	6	NKG7	
0	2.026542	0.367	0.054	0	6	GNLY	
0	1.961304	0.75	0.068	0	6	GZMB	
0	1.917466	0.468	0.058	0	6	GZMA	
0	1.731511	0.423	0.009	0	6	GZMH	
0	1.699852	0.213	0.002	0	6	GZMK	
0	1.634118	0.817	0.134	0	6	CTSW	
0	1.564998	0.668	0.07	0	6	CD8A	
0	1.446734	0.908	0.224	0	6	CD3D	
0	1.319366	0.723	0.05	0	7	TK1	KC-4
0	1.243987	0.873	0.3	0	7	STMN1	
0	1.226331	0.915	0.24	0	7	PCNA	
0	1.151753	0.989	0.785	0	7	H2AFZ	
0	1.1299	0.98	0.574	0	7	MT1X	
0	1.091659	0.712	0.051	0	7	PCLAF	
0	1.058725	0.992	0.793	0	7	TUBA1B	
0	1.014724	0.893	0.411	0	7	HMGB2	
0	0.990478	0.867	0.245	0	7	CKS1B	
2.1E-216	1.567707	0.819	0.499	7.1E-212	7	HIST1H4C	
0	2.021471	0.971	0.296	0	8	PTTG1	KC-5
0	1.71031	0.91	0.067	0	8	CENPF	
0	1.492718	0.811	0.033	0	8	CDC20	

0	1.366117	0.804	0.045	0	8	CDKN3	
0	1.355426	0.723	0.039	0	8	CCNB1	
0	1.328533	0.995	0.452	0	8	KRT5	
0	1.261056	0.986	0.654	0	8	MT2A	
0	1.241981	0.978	0.684	0	8	HMG2	
0	1.234653	0.825	0.298	0	8	KRT17	
0	1.211345	0.891	0.185	0	8	NMU	
0	2.129638	0.735	0.153	0	9	TNFRSF4	TREG
0	1.747588	0.77	0.041	0	9	CD27	
0	1.60785	0.741	0.074	0	9	TIGIT	
0	1.601941	0.862	0.224	0	9	TRBC2	
0	1.598999	0.605	0.06	0	9	IL2RA	
0	1.568397	0.953	0.276	0	9	TRAC	
0	1.556205	0.58	0.149	0	9	TRBC1	
0	1.529469	0.624	0.033	0	9	CTLA4	
0	1.443848	0.996	0.296	0	9	IL32	
4.8E-291	1.344107	0.795	0.417	1.6E-286	9	TNFRSF18	
0	3.194721	0.862	0.016	0	10	FCGBP	LC
0	3.167698	0.945	0.065	0	10	HLA-DQB2	
0	2.568222	0.882	0.096	0	10	S100B	
0	2.553904	0.941	0.028	0	10	CD207	
0	2.486175	0.984	0.126	0	10	FCER1A	
0	2.393265	0.982	0.091	0	10	CD1A	
0	2.067857	0.835	0.178	0	10	HLA-DRB5	
0	1.939942	0.999	0.452	0	10	CD74	
0	1.873083	0.982	0.147	0	10	CPVL	
0	1.781478	0.798	0.127	0	10	PTGS2	
0	2.796156	0.98	0.155	0	11	CTSW	TC-6
0	2.472756	0.842	0.024	0	11	XCL1	
0	2.437623	0.8	0.091	0	11	GZMB	

0	2.425002	0.766	0.057	0	11	GNLY	
0	2.252579	0.821	0.016	0	11	SPINK2	
0	2.132071	0.739	0.01	0	11	XCL2	
0	1.912208	0.981	0.416	0	11	TNFRSF18	
0	1.809806	0.783	0.024	0	11	KRT86	
0	1.585911	0.673	0.008	0	11	KRT81	
0	1.582289	0.66	0.013	0	11	MB	
0	2.442272	0.819	0.077	0	12	IL13	TC-3
0	1.862581	0.665	0.083	0	12	KLRB1	
0	1.587125	0.936	0.255	0	12	NBAS	
0	1.525844	0.906	0.286	0	12	CD69	
0	1.476995	0.989	0.35	0	12	ALOX5AP	
0	1.464979	0.983	0.304	0	12	IL32	
0	1.356693	0.662	0.011	0	12	IL17RB	
0	1.315323	0.476	0.02	0	12	HPGDS	
0	1.278708	0.975	0.339	0	12	SAMSN1	
1E-271	1.353736	0.862	0.289	3.5E-267	12	BIRC3	
1.9E-193	1.845674	0.987	0.974	6.2E-189	13	MALAT1	Other
9E-136	1.31602	0.946	0.95	3E-131	13	MT-ND3	
1.38E-10	1.836122	0.033	0.156	4.62E-06	13	SLPI	
3.14E-08	1.427143	0.401	0.414	0.001052	13	EML4	
9.49E-08	1.260233	0.462	0.545	0.003184	13	MTRNR2L12	
3.98E-07	1.23667	0.408	0.443	0.013365	13	STK4	
9.22E-05	1.149208	0.077	0.162	1	13	LTC4S	
0.000136	1.192519	0.314	0.299	1	13	BIRC3	
0.000562	1.19171	0.426	0.562	1	13	N4BP2L2	
0.004664	1.163214	0.293	0.313	1	13	PCSK7	
0	3.956284	0.425	0.007	0	14	CCL1	TC-4
0	2.296023	0.748	0.085	0	14	IL13	
0	1.891933	0.568	0.024	0	14	CRTAM	

0	1.805192	0.51	0.04	0	14	CSF2	
3.2E-242	1.455839	0.844	0.164	1.1E-237	14	TNFRSF4	
6E-166	1.69012	0.354	0.038	2E-161	14	XCL1	
7.6E-164	1.510613	0.959	0.423	2.6E-159	14	TNFRSF18	
8E-119	2.11518	0.354	0.051	2.7E-114	14	IL22	
7.7E-106	1.216826	0.949	0.57	2.6E-101	14	REL	
1.17E-34	1.303509	0.327	0.104	3.93E-30	14	GZMB	
0	3.72898	0.926	0.029	0	15	CCL22	DC-2
0	3.226053	0.643	0.059	0	15	CCL17	
0	2.562021	0.97	0.017	0	15	CCR7	
0	2.310431	0.926	0.169	0	15	BASP1	
0	2.106668	0.807	0.071	0	15	CD1B	
0	2.100381	0.974	0.033	0	15	LAMP3	
2.6E-273	2.427541	0.974	0.248	8.6E-269	15	FSCN1	
8.6E-233	2.2303	0.981	0.294	2.9E-228	15	BIRC3	
1.1E-222	2.23499	0.974	0.284	3.6E-218	15	MARCKSL1	
5.5E-152	2.501577	0.996	0.892	1.8E-147	15	TXN	
0	1.721159	0.868	0.08	0	16	TYMS	TC-5
0	1.498192	0.756	0.072	0	16	MKI67	
0	1.368862	0.778	0.074	0	16	PCLAF	
2.7E-201	2.163354	0.977	0.32	9.2E-197	16	STMN1	
4.1E-135	1.58468	0.951	0.428	1.4E-130	16	HMGB2	
1.4E-120	1.364739	0.936	0.489	4.6E-116	16	DUT	
1.3E-101	1.654569	0.951	0.594	4.41E-97	16	TUBB	
3.9E-97	1.454479	0.996	0.8	1.31E-92	16	TUBA1B	
7.47E-56	1.396388	0.109	0.01	2.5E-51	16	CCL1	
1.41E-44	1.674502	0.82	0.51	4.72E-40	16	HIST1H4C	
0	2.31566	0.356	0.001	0	17	JCHAIN	DC-3
0	2.267179	0.908	0.046	0	17	IRF8	
0	1.983807	0.331	0.002	0	17	IGKC	

9.1E-252	1.774702	0.742	0.074	3.1E-247	17	DNASE1L3	
1.5E-131	2.078564	0.81	0.162	5.1E-127	17	CPVL	
1.3E-108	2.060167	0.939	0.264	4.3E-104	17	LTB	
3E-108	2.00096	1	0.377	1E-103	17	HLA-DPB1	
1.3E-104	1.939341	1	0.405	4.5E-100	17	HLA-DPA1	
1.73E-96	1.96459	1	0.462	5.79E-92	17	CD74	
3.97E-67	1.841262	0.969	0.627	1.33E-62	17	CST3	

TABLES S5-S10: Please see separate Excel files

Table S5. Differential gene expression analysis within cell types of scRNA-seq analyses; comparison of healthy controls versus AD untreated. Defined by $\log FCH > |0.25|$ and adjusted p-value < 0.05, using logistic regression and Bonferroni correction. AD: Atopic dermatitis; HC: Healthy controls.

Table S6. Differential gene expression analysis within cell types of scRNA-seq analyses; comparison of 16 weeks dupilumab-treated AD versus AD untreated. Defined by $\log FCH > |0.25|$ and adjusted p-value < 0.05, using logistic regression and Bonferroni correction. AD: Atopic dermatitis; 16w: 16 weeks of dupilumab treatment in AD.

Table S7. Differential gene expression analysis within cell types of scRNA-seq analyses; comparison of 1 year dupilumab-treated AD versus AD untreated. Defined by $\log FCH > |0.25|$ and adjusted p-value < 0.05, using logistic regression and Bonferroni correction. AD: Atopic dermatitis; 1y: 1 year of dupilumab treatment in AD.

Table S8. Differential gene expression analysis within cell types of scRNA-seq analyses; comparison of 16 weeks of dupilumab-treated AD versus healthy controls. Defined by $\log FCH > |0.25|$ and adjusted p-value < 0.05, using logistic regression and Bonferroni correction. AD: Atopic dermatitis; HC: Healthy controls; 16w: 16 weeks of dupilumab treatment in AD.

Table S9. Differential gene expression analysis within cell types of scRNA-seq analyses; comparison of 1 year of dupilumab-treated AD versus healthy controls. Defined by $\log FCH > |0.25|$ and adjusted p-value < 0.05, using logistic regression and Bonferroni correction. AD: Atopic dermatitis; HC: Healthy controls; 1y: 1 year of dupilumab treatment in AD.

Table S10. Differential gene expression analysis within cell types of scRNA-seq analyses; comparison of 1 year of dupilumab-treated AD versus 16 weeks of dupilumab-treated AD. Defined by $\log FCH > |0.25|$ and adjusted p-value < 0.05, using logistic regression and Bonferroni correction. Dupi16w: 16 weeks of dupilumab treatment in AD; Dupi1y: 1 year of dupilumab treatment in AD.

Table S11. Percentages of *IL13*⁺, *IL26*⁺, *IL22*⁺, and *IFNG*⁺ cells, as percentages of all T2 cells (*IL13*⁺), T17 cells (*IL26*⁺), T22 cells (*IL22*⁺), and T1 cells (*IFNG*⁺). As assessed by scRNA-seq, demonstrating frequencies of cells showing double positivity for these cytokines. AD: Atopic dermatitis.

IL13+ CD3D+ population	AD untreated	AD 16 weeks dupilumab	AD 1 year dupilumab	Healthy controls
IL26+ cells (% of all IL13+ CD3D+ cells)	41.86	44.64	40.75	13.18
IL22+ (% of all IL13+ CD3D+ cells)	38.72	43.66	35.98	13.93
IFNG+ (% of all IL13+ CD3D+ cells)	4.58	5.76	3.03	2.75
IL26+ CD3D+ population	AD untreated	AD 16 weeks dupilumab	AD 1 year dupilumab	Healthy controls
IL13+ (% of all IL26+ CD3D+ cells)	50.70	64.28	76.20	32.55
IL22+ (% of all IL26+ CD3D+ cells)	39.02	44.60	47.08	13.40
IFNG+ (% of all IL26+ CD3D+ cells)	5.12	5.02	2.53	7.45
IL22+ CD3D+ population	AD untreated	AD 16 weeks dupilumab	AD 1 year dupilumab	Healthy controls
IL13+ (% of all IL22+ CD3D+ cells)	58.88	61.26	72.43	36.95
IL26+ (% of all IL22+ CD3D+ cells)	48.76	45.76	49.75	20.35
IFNG+ (% of all IL22+ CD3D+ cells)	4.08	5.90	2.20	0.00
IFNG+ CD3D+ population	AD untreated	AD 16 weeks dupilumab	AD 1 year dupilumab	Healthy controls
IL13+ (% of all IFNG+ CD3D+ cells)	26.08	36.38	35.30	5.05
IL26+ (% of all IFNG+ CD3D+ cells)	26.43	20.76	18.65	8.18
IL22+ (% of all IFNG+ CD3D+ cells)	19.48	19.74	9.75	0.00

TABLE S12: Please see separate Excel file

Table S12. Differential protein expression of proteomic multiplex assays from suction blister fluid samples. Using a linear mixed model and Benjamini-Hochberg correction. AD: Atopic dermatitis; D16w: 16 weeks of dupilumab-treated AD; D1y: one year of dupilumab-treated AD; HC: Healthy controls.

Table S13. Baseline characteristics of patients used for serum sample analyses (multiplex proteomic assays). At baseline (untreated AD), 16 weeks and 1 year of dupilumab treatment, as well as healthy control (HC) individuals. EASI: Eczema Area and Severity Index; n.a. not available.

Patient ID	Sex	Race	Age (years)	Condition	EASI at baseline (untreated AD)	EASI at 16 weeks of dupilumab treatment	EASI at 1 year of dupilumab treatment
S2	male	Caucasian	36	Atopic dermatitis	44.7	14.9	10.8
S3	male	Caucasian	35	Atopic dermatitis	38.4	2.9	0.1
S4	female	Caucasian	46	Atopic dermatitis	27.4	5.4	n.a.
S5	female	Caucasian	21	Atopic dermatitis	38.6	2.4	n.a.
S6	male	Caucasian	33	Atopic dermatitis	52.4	7.5	9.8
S7	male	Caucasian	42	Atopic dermatitis	41.6	4.8	1.1
S8	male	Caucasian	33	Atopic dermatitis	36.2	10.8	3.0
S9	male	Caucasian	47	Atopic dermatitis	44.5	8.2	n.a.
S11	female	Caucasian	56	Atopic dermatitis	19.9	n.a.	n.a.
S12	female	Caucasian	80	Atopic dermatitis	21.2	17.3	5.6
S13	female	Caucasian	56	Atopic dermatitis	52.0	22.0	n.a.
S14	male	Caucasian	20	Atopic dermatitis	34.2	10.1	n.a.
S15	male	Caucasian	51	Atopic dermatitis	42.2	1.2	0.1
S16	female	Caucasian	35	Atopic dermatitis	19.9	0.6	n.a.
S17	male	Caucasian	52	Atopic dermatitis	43.6	30.3	n.a.
S18	male	Caucasian	32	Atopic dermatitis	26.8	4.2	1.5
S19	male	Caucasian	37	Atopic dermatitis	31.8	n.a.	13.7
S20	male	Caucasian	29	Atopic dermatitis	29.4	6.0	n.a.
S21	male	Caucasian	32	Atopic dermatitis	47.3	25.4	n.a.
S22	female	Caucasian	33	Atopic dermatitis	20.3	2.2	7.7

S23	male	Caucasian	26	Atopic dermatitis	48.0	15.9	10.8
S24	female	Caucasian	61	Atopic dermatitis	44.8	2.6	n.a.
S26	female	Caucasian	30	Atopic dermatitis	31.8	9.2	2.6
G7	male	Caucasian	47	Healthy control	n.a.	n.a.	n.a.
G29	male	Caucasian	48	Healthy control	n.a.	n.a.	n.a.
G25	male	Caucasian	29	Healthy control	n.a.	n.a.	n.a.
G20	male	Caucasian	27	Healthy control	n.a.	n.a.	n.a.
G18	female	Caucasian	27	Healthy control	n.a.	n.a.	n.a.

TABLE S14-S15: Please see separate Excel file

Table S14. Differential protein expression of proteomic multiplex assays from serum samples. Using an unpaired double-sided t-test and Benjamini-Hochberg correction. AD: Atopic dermatitis; D16w: 16 weeks of dupilumab-treated AD; D1y: one year of dupilumab-treated AD; HC: healthy controls.

Table S15. Raw data file (Excel spreadsheet).